

Paul Schulte

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EDUCATION

RWTH Aachen University

M.Sc. Computer Science; GPA: 1.2 (1.0 best)

Oct 2025 – Present

Aachen, Germany

- **Focus:** Machine Learning, Optimization, Algorithms, and Graph Theory

RWTH Aachen University

B.Sc. Computer Science; GPA: 1.1 with distinction (1.0 best)

Oct 2022 – Aug 2025

Aachen, Germany

- **Dean's List:** Top 5% of cohort across all semesters; completed degree in 5 of 6 semesters
- **Bachelor Thesis:** *Action Classification Through Dynamic Scene Graph Generation from Image Sequences* (Grade: 1.0) 

Aalto University

Semester Abroad, Computer Science; GPA: 5.0 / 5.0

Aug 2024 – Dec 2024

Espoo, Finland

- **Coursework:** Competitive Programming, Distributed Algorithms, Graph Theory

RESEARCH EXPERIENCE

RWTH Aachen University, Chair for Machine Learning and Inference (i6)

Graduate Research Assistant

Aachen, Germany

Sep 2025 – Present

- Researching self-supervised learning of discrete relational structures from image sequences
- Extended thesis work on dynamic scene graph generation for action classification
- Equal-contribution first author of an oral GenPlan Workshop paper at ICAPS 2026

Robotics Institute, Carnegie Mellon University

Undergraduate Research Intern (RISE Worldwide)

Pittsburgh, PA, USA

May 2024 – Sep 2024

- Developed an MPC framework with Control Barrier Functions for safety-critical trajectory planning
- Implemented real-time planning in JAX for CPU-based F1TENTH hardware, reducing reliance on GPU-heavy approaches
- Co-authored the USDOT/Safety21 final report *Risk-Aware Warning and Control for Interactive Traffic Safety* 

PROFESSIONAL EXPERIENCE

Siemens Logistics

Software Engineering Working Student

Düsseldorf, Germany

Feb 2023 – Nov 2024

- Designed and implemented C# features for an airport baggage handling system planning tool
- Optimized computationally intensive planning components, reducing runtime by up to 80%

SELECTED PROJECTS

Math Exercise Generator | *JavaScript, HTML, Vue.js*

2023

- Built a web-based linear algebra exercise generator with automatic solution computation

Neuroevolution of Fixed Topologies | *Python, Reinforcement Learning*

2021

- Implemented neural networks and evolutionary algorithms from scratch
- Applied methods to OpenAI Gym environments and documented results in a technical report

PUBLICATIONS

Ulzhalgas Rakhman*, **Paul Schulte***, Hector Geffner. *Learning Lifted Symbolic Planning Domains from Image Sequences (VISA)*. GenPlan Workshop at ICAPS 2026. Oral presentation. *Equal contribution.

TECHNICAL SKILLS

Programming: Python, C++, Java, C#, JavaScript

ML / Scientific Computing: JAX, NumPy, Machine Learning, Optimization

Tools: Linux, Git, Docker, L^AT_EX

Languages: German (native), English (C1)

HONORS & DISTINCTIONS

DAAD RISE Worldwide Scholarship (2024) | **German Mathematical Society** student member (2022) | **German Physical Society** student member (2022)